

Arden Tsang

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EDUCATION

University of Cambridge

MASt in Astrophysics

Expected graduation: Jun 2027

Cambridge, UK

- Selected courses: QFT, GR, Black Holes, Field Theory in Cosmology, GWs & Numerical Relativity

King's College London (KCL)

Sep 2023 - Jul 2026

BSc Physics with Theoretical Physics

London, UK

- First Class Honours** every year, Ranked **Top 5%** of year. Overall weighted average: **79%** (final year **81%**)
- King's Research Award.**
- Layton Science Research Award.** "for the best promise of aptitude and genius for original scientific work."
- Relativistic Astrophysics[91], QM II[89], Condensed Matter[84], EM[83], Particle Physics[81], QM I[81], Medical Imaging[81], Optics[79], Math in Physics[79], Theoretical Physics[77] & Symmetry[76], Stat Mech[75].

RESEARCH & EXPERIENCE

COSMOS INSTITUTE GRANT (WITH FIRE); GROUNDED VERIFICATION BENCHMARK

Aug 2026 – Present

Eden/Independent

London, UK

- Built a benchmark on an adjudicated dispute over an autonomous laboratory's synthesis claims with expert-derived per-claim targets and a public scorer that evaluates outside submissions. [[Git](#)]
- Designed a verification layer in which a model proposes checks and an independent kernel re-establishes every verdict; reached 55/55 stage-one targets against 41 for the strongest unlayered baseline.

CCP5 RESEARCH FELLOW; THEORETICAL & COMPUTATIONAL MATERIALS PHYSICS

Dec 2025 – Present

KCL De Tomas Group

London, UK

- Awarded EPSRC-funded CCP5** (Condensed Matter Simulation Project) Bursary for full-time summer extension.
- Developed and characterised a hybrid PSO–DFT (GFN2-xTB/PBE+VV10) workflow for carbon-nanocluster structure prediction; benchmarked inertias, recovered C_{3-20} isomers and converged a C_{20} fullerene.
- Built HPC pipelines in Bash for GPU-accelerated optimiser tuning; authored READMEs for reproducibility. [[Git](#)]
- First authoring** a manuscript on this work; submission targeted for September 2026.
- Curated benchmark datasets to validate against GAP-20/DFT, bridging ab initio accuracy with ML-IP scalability.
- Extending to compare simulated IR spectra against astrophysical molecular cloud observations.
- Working on PSO to C_8O_2 , C_8S_2 & initiating interpretability work on random-forest feature attribution; automated kinetic-state discovery via PySoftK. Supervised by Dr Carla De Tomas, Senior Lecturer in Net Zero.

PARTICLE PHYSICS & MACHINE LEARNING RESEARCH ASSISTANT

Oct 2025 – Present

KCL Experimental Particle and Astroparticle group (EPAP)

London, UK

- Co-authored paper:** "Physics at the Edge: Benchmarking Quantisation Techniques and the EdgeTPU for Neutrino Interaction Recognition". ([arXiv:2603.24607](https://arxiv.org/abs/2603.24607), submitted to EPJC)
- Presented poster at the AI for Science Conference, Alan Turing Institute, March 2026.
- Leading work (build, train, quantise) in GNNs (GATs & GCNs) & architecting ground up EGNNs to run on Edge.
- Running reconstruction and clustering with Pandora; writing ROOT files for the group's set-transformer models.
- Building mechanistic interpretability work on set-transformer and graph-attention models.
- Benchmarking quantum machine learning circuits across quantum hardware architectures on accuracy, latency and energy cost, analysing all-to-all circuits compilation.
- Trained and quantized CNNs (PTQ & QAT) with GENIE generated LArTPC neutrino datasets to support possible low latency, sustainable AI fast triggers. Supervised by Dr Stefano Vergani, Postdoctoral Research Associate.

PHOTOVOLTAICS UNDERGRADUATE RESEARCH ASSISTANT

Sep 2025 – Jun 2026

KCL Functional Materials & Devices Laboratory

London, UK

- Integrated CAD: Fusion360 & Inkscape with V4.75 CO₂ laser to nanostructure perovskite for spectral filtering.
- Investigated dichroic filters, perovskite for agrivoltaics; Supervised by Dr Jay Patel, Functional Photonics Lecturer.

ARTIFICIAL INTELLIGENCE UNDERGRADUATE SUMMER RESEARCH FELLOW

Jul 2025 – Sep 2025

HKUST Industrial Engineering and Decision Analytics (IEDA)

Hong Kong

- Designed and programmed scheduling algorithms for NPU architectures to optimize inference speed.
- Learned transformer theory & curated datasets to fine-tune LLMs with mlx to test physical derivation recovery.
- Interviewed published researchers on AI safety. Supervised by Dr Jiheng Zhang, Professor of IEDA. & Mathematics.

QUANTUM CHEMISTRY KING'S UNDERGRADUATE RESEARCH FELLOW (KURF)

Jun 2025 – Jul 2025

KCL Physical Chemistry (Sanz) Group

London, UK

- Successful identification of the lowest-energy isomer of limonene–(SO₂)₂ cluster via FT-CPMW spectroscopy.
- Employed Quantum Chemistry suite: DFT on the HPC via CLI; Spectral assignment, & NCI, NBO Chemical analysis.

- Awarded paid research fellowship; Supervised by Dr Maria Eugenia Sanz, Reader in Physical Chemistry.

ACTUARIAL INTERN

Jul 2024 – Sep 2024

AIA

Hong Kong

- Studied credit risk algorithms & created market projections, executed data analysis & crafted quarterly report.
- Documented linear algebra legacy code in Python and VBA from Excel to senior colleagues.

PUBLICATIONS, POSTERS & CONFERENCES

- Tsang, A. *Empirical Verificational Complexity: A Research Programme for Scalable Oversight and Steering in Autonomous Laboratories*. Manuscript under review, 2026.
- Vergani, S.*, **Tsang, A.**, et al. *Physics at the Edge: Benchmarking Quantisation Techniques and the Edge TPU for Neutrino Interaction Recognition*. **AI for Science Conference, Alan Turing Institute**, London, UK. Mar 2026
- **Tsang, A.**, Kuchmina, S., de Tomas, C., et al. *Simulating Carbon Nanoclusters using ML-IP & Particle Swarm Optimisation*. **London Centre for Nanotechnology Symposium**; BSc Thesis Poster Session, London, UK, Apr 2026
- **Tsang, A.**, Sanz, M. E. *Atmospheric Clusters: Structure and Interactions*. Academic poster submitted for the King's Research Award, **King's Undergraduate Research Fellowship (KURF)**, London, UK, Oct 2025

POSITIONS OF RESPONSIBILITY

Secret Loyalties Hackathon, Apart

(London, UK) Jul 2026

- Built mech interp suite (logit/j-lens, causal attention ablations) to search for 'secret loyalties' in fine-tuned LLMs

Cosmology Hackathon, KCL Department of Physics

(London, UK) Jun 2026

- Built a ML pipeline to infer galaxy redshifts from 5-band SDSS photometry using GBTs and MLPs; RMSE $\approx$ 0.052.

Staff Writer, King's International Security Journal (KISJ)

(London, UK) Sep 2025 – Present

- Sole author on AI's authoritarian tendencies, misalignment, collapse of information scarcity & macro-economy

Engineering, Philosophy Lead, KCL Rocketry Society

(London, UK) Sep 2024 – Feb 2026

- Showcased technical exhibition & facilitated physical rocket construction utilizing Fusion360 for modeling.

Elected Head, SMUS Physics Club

(Victoria, BC, Canada) Sep 2022 – Jun 2023

- Executed 15+ experiments, tested hypotheses, argued methods & results for theory; Hosted lectures on QM.

Co-Founder & Head, SMUS Engineering Club

(Victoria, BC, Canada) Sep 2021 – Jun 2022

- Recruited 80+; Lectured; Directed bridge competition; Taught Inkscape to design & compete with printed car.

Physics Competitions; [IOP/IAPS], [CAP/IQC]

(Swansea, UK); (Victoria, BC, Canada) Sep 2021 – Feb 2025

- **PLANCKS**: Winner of promotional event, competed & networked across UK universities in theory.

- **CaYPT**: Led team of 5 to design experiments, debated methods, results, & theories with schools across Canada.

STEM Tutor; Various Institutions

(Victoria, BC, Canada); (London, UK) Sep 2021 – Present

- Instructed 15+ students in AP & IB Physics, AP Calc, Chem, Stat; across English & Cantonese, remote & in-person

COURSES & TRAINING

TECHNICAL AI SAFETY

Jul 2026

BlueDot Impact

Remote

- Alignment & RLHF, mech. interp., evaluations & red-teaming, AI control, & scalable oversight; capstone project.

FRONTIER AI GOVERNANCE

Jul 2026

BlueDot Impact

Remote

- Compute governance, safety standards, liability, & international coordination; analysis of AI policy proposals.

TECHNICAL SKILLS

- **Languages**: Native English, Native Cantonese, Fluent Mandarin, Limited Working Proficiency Spanish.
- **Programming & Computation**: Python (PySCF, gpu4pyscf, PyTorch, TensorFlow, Keras, MLX, SciPy, Pandas, Numpy, Matplotlib), Bash, MATLAB; Java; VBA; Swift;
- **HPC & Systems**: Linux/Unix CLI (MobaXterm, MacOS, Windows), Schedulers, GPU-accelerated workflows.
- **Software & Tools**: $\LaTeX$; Git; Quantum Chemistry: PGOPHER, Pickett, Gaussian, Chimera, Multiwfn, OVITO, NetworkX; CAD: Fusion360, Inkscape, Prusa; Sketchup.
- **Machine Learning & Numerical Methods**: LLM fine-tuning (MLX); GNNs; CNNs; Quantization: (Post-Training-Quantization, Quantization-Aware-Training), Stochastic Optimisation (PSO, Simulated Annealing); ODEs/PDEs (Mckenna's model, Taylor's, Tacoma); Packing (NPU); QM, MRI and ultrasound theory & visualisation.
- **Mathematical Physics**: Lie Algebras & Representation Theory; Analytical Mechanics; Quantum, Relativistic & Statistical Mechanics; Tensor Calculus; Axiomatic Set Theory.
- **Experimental Physics**: Optics & Photonics: Interferometry, Laser alignment, Spectroscopy, Polarimetry. Instrumentation: CO₂ lasers, Optical bench, Oscilloscopes, Helmholtz coils, NMR Spectrometers, Circuit design.